Electronic Dice Game using LPC2129 & 16×2 LCD

The Electronic Dice Game uses an LPC2129 microcontroller and a 16×2 LCD to replicate a traditional die for board games. It generates numbers from 1 to 6 and displays them on the LCD, creating a simple, interactive gaming experience. The system includes an LPC2148 ARM7 microcontroller, a 16×2 LCD, and two push-button switches—one for each player. While a player holds a switch, the microcontroller rapidly cycles through numbers to simulate a rolling die. When the switch is released, the number shown becomes the final dice value.

The project is developed using Embedded C and demonstrates key concepts such as microcontroller programming, LCD interfacing, switch interfacing, GPIO control, pseudo-random number generation, and real-time user interaction. It serves as an effective educational tool for understanding embedded system design and hardware-software integration. The Electronic Dice Game is portable, easy to use, and eliminates the need for a physical dice, making it suitable for laboratory experiments, mini-project demonstrations, and gaming applications.

Future Scope:

Future enhancements include support for 4-player and 6-player modes based on user requirements, score tracking, automatic winner declaration, buzzer and LED indications, graphical displays, and wireless connectivity. These improvements will make the system more interactive, scalable, and suitable for a wider range of gaming applications.

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